A HAUSDORFF SPACE WHICH IS THE CLOSURE OF THE IMAGE OF A CONTINUOUS MAP HAS CARDINALITY AT MOST THE DOUBLE POWER SET OF THE SOURCE

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Allowed standard prerequisites (editorial).

- The notion of a dense subset and the Hausdorff separation axiom, together with the cardinality conventions used in this section. Set up in the author's section on the Stone-Cech compactification, where the counted statement occurs. Fixes the hypothesis on the image and the conclusion about the cardinality of the target.

Lemma 0.1. *Let $f : X \to Y$ be a continuous map of topological spaces. Assume that $f(X)$ is dense in Y and that Y is Hausdorff. Then the cardinality of Y is at most the cardinality of $P(P(X))$ where P is the power set operation.*

Proof. Let $S = f(X) \subset Y$. Let $\mathcal{D}$ be the set of all closed domains of Y, i.e., subsets $D \subset Y$ which equal the closure of its interior. Note that the closure of an open subset of Y is a closed domain. For $y \in Y$ consider the set

$$I_y = \{T \subset S \mid \text{ there exists } D \in \mathcal{D} \text{ with } T = S \cap D \text{ and } y \in D\}.$$

Since S is dense in Y for every closed domain D we see that $S \cap D$ is dense in D. Hence, if $D \cap S = D' \cap S$ for $D, D' \in \mathcal{D}$, then $D = D'$. Thus $I_y = I_{y'}$ implies that $y = y'$ because the Hausdorff condition assures us that we can find a closed domain containing y but not y'. The result follows. □

Date: stacks-project:topology.tex:local-snapshot-20261012; CC BY 4.0.